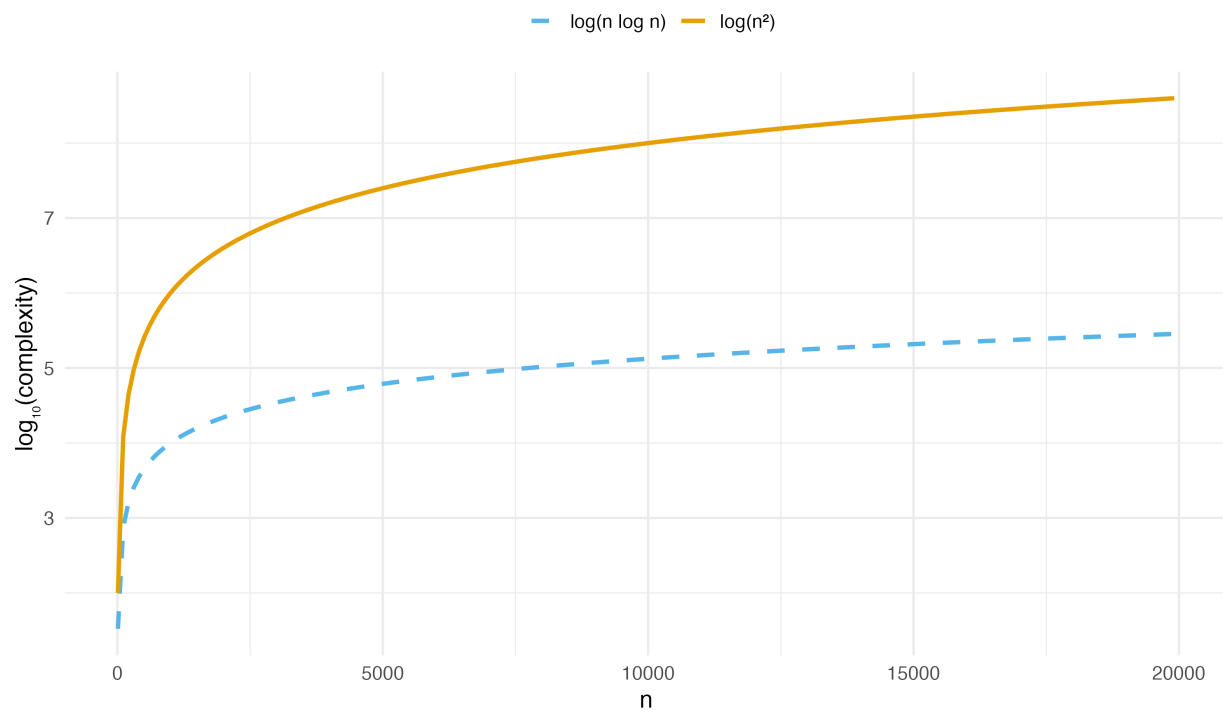


Search Algorithms Performance Analysis Report

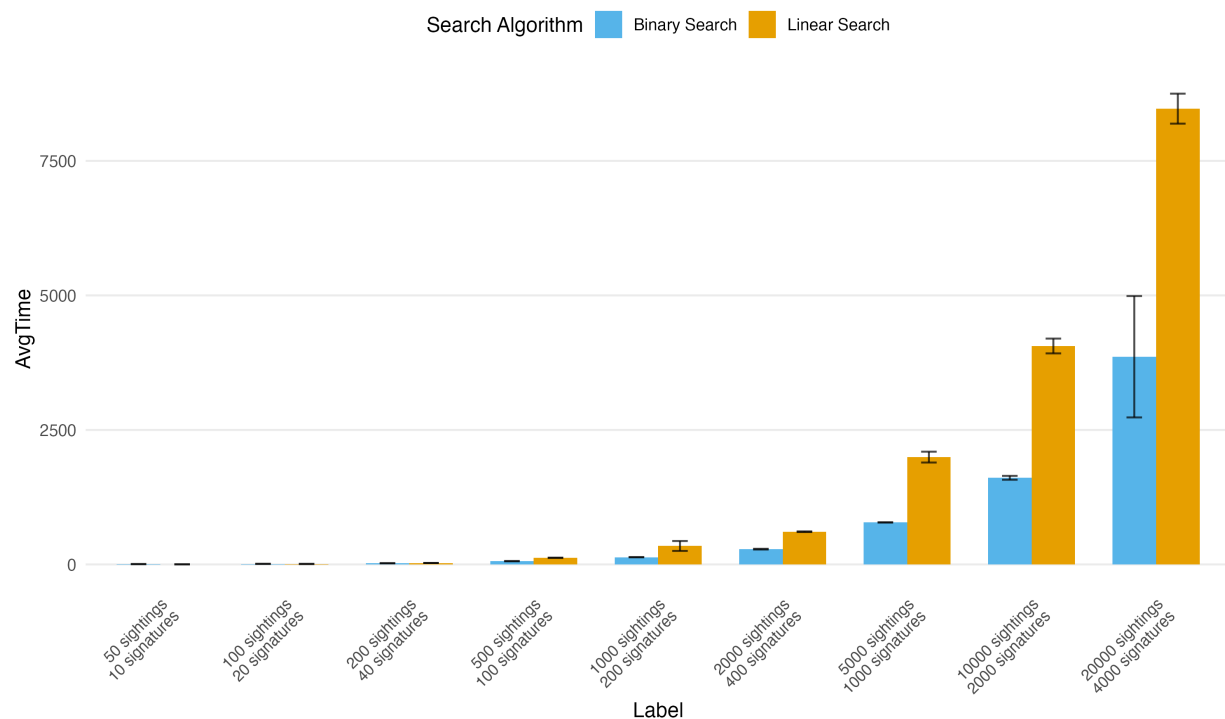
Based on the code we can analyze Search Algorithms time complexity:

Linear Search: $O(n*m)$

Binary Search: $O(n \log n_{\text{sort}} + m \log n_{\text{search}})$



SIGHTINGS	SIGNATURES	LINEAR	BINARY	SPEEDUP
20000	4000	8470.6	3860.31	2.19
10000	2000	4059.97	1610.33	2.52
5000	1000	1994.95	781.24	2.55
2000	400	608.32	284.25	2.14
1000	200	343.57	134.79	2.55
500	100	123.93	61.33	2.02
200	40	27.18	22.5	1.21
100	20	9.29	10.86	0.86
50	10	2.24	6.72	0.33



Based on the test if the n is less than 200, Binary Search might slower than Linear Search.

Why Binary Search Doesn't Achieve Theoretical Advantage Immediately

might be because sorting : One-time $O(n \log n)$ cost is significant for small datasets

Test Environment

Hardware: - Processor: Apple Silicon (ARM64) - Memory: 16G

Software: - OS: macOS 26 (Darwin 25.2.0) - Compiler: g++ (Apple clang) - Flags:

-std=c++11 -Wall -Werror

Data Collection: - Test runs: 3 per configuration - Total configurations: 27 - Total measurements: 54